# Case Study-1

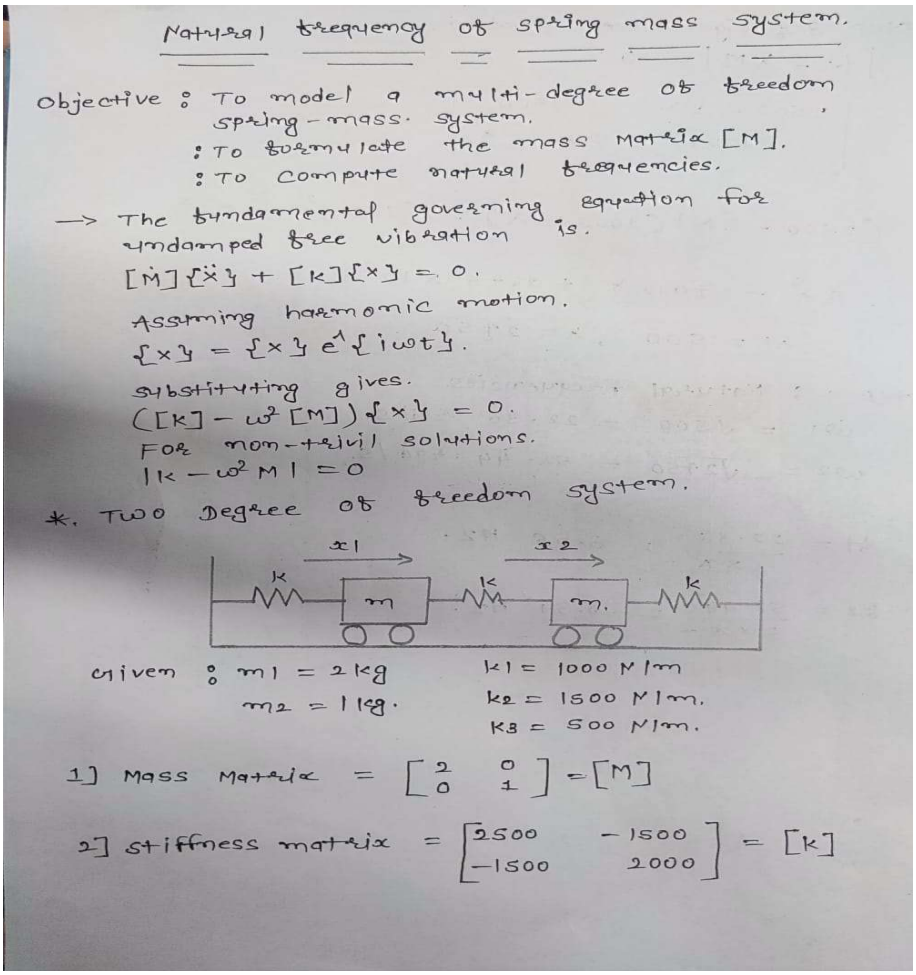
# Natural Frequency of Spring Mass System

- Theory :-

Vibration is a common phenomenon in mechanical and structural systems such as bridges, engines, machines, and vehicles. Every mechanical system has a natural frequency, and when external excitation matches this frequency, resonance occurs, leading to excessive vibration and possible failure.

The spring–mass system is the simplest model used to study vibration behavior. By forming mass matrix [M] and stiffness matrix [K], and solving the eigenvalue problem, natural frequencies and mode shapes of the system can be determined.

This case study focuses on estimating natural frequencies using matrix and eigenvalue methods.



Natural frequency of spring mass system.

Objective : To model a multi-degree of freedom spring-mass system.
: To formulate the mass matrix [M].
: To compute natural frequencies.

→ The fundamental governing equation for undamped free vibration is.

$[M]\{\ddot{x}\} + [K]\{x\} = 0.$

Assuming harmonic motion.

$\{x\} = \{x\} e^{i}\{i\omega t\}.$

substituting gives.

$([K] - \omega^2 [M])\{x\} = 0.$

For non-trivil solutions.

$|K - \omega^2 M| = 0$

*. Two Degree of freedom system.



Given : $m1 = 2kg$ $k1 = 1000 N/m$
$m2 = 1kg.$ $k2 = 1500 N/m.$
$k3 = 500 N/m.$

1] Mass Matrix $= \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = [M]$

2] stiffness matrix $= \begin{bmatrix} 2500 & -1500 \\ -1500 & 2000 \end{bmatrix} = [k]$

Step 3 : Eigen value Equation.

$$|K - \lambda M| = 0.$$

$$\begin{vmatrix} 2500 - 2\lambda & -1500 \\ -1500 & 2000 - \lambda \end{vmatrix} = 0.$$

$$(2500 - 2\lambda)(2000 - \lambda) - (1500)^2 = 0.$$

$$2\lambda^2 - 7000\lambda + 2.75 \times 10^6 = 0.$$

$$\boxed{\lambda 1 = 500 \quad , \quad \lambda = 2750}$$

Step 4 : Natural frequencies.

$$\omega 1 = \sqrt{500} = 22.36 \text{ rad/s.}$$

$$\omega 2 = \sqrt{2750} = 52.44 \text{ rad/s.}$$

$$f1 = \frac{22.36}{2 \times \pi} = 3.56 \text{ Hz.}$$

$$f2 = \frac{52.44}{2 \times \pi} = 8.34 \text{ Hz.}$$

## Project Overview

This project focuses on determining the natural frequencies and mode shapes of a two-degree-of-freedom (2-DOF) spring–mass system using mass and stiffness matrix formulation and eigenvalue analysis in MATLAB.

The natural frequencies are obtained when:

• The dynamic equilibrium equation  is satisfied

• The determinant condition  produces non-trivial solutions.[M]{\ddot{x}} + [K]{x}} = 0

This project demonstrates how matrix formulation and computational tools can be used to analyze vibration behavior of mechanical systems efficiently using MATLAB.

→ Objective

The main objectives of this project are:

• To understand the concept of free vibration in multi-degree-of-freedom systems

• To formulate mass and stiffness matrices

• To apply eigenvalue analysis

• To compute natural frequencies and mode shapes

## Problem Description

A two-degree-of-freedom spring–mass system consisting of:

• Two masses

• Three linear springs

• Fixed supports is analyzed to determine its natural frequencies and vibration response.

The system parameters used are:

• $m_1$ = 4 kg, $m_2$ = 2 kg

• $k_1$ = 2000 N/m, $k_2$ = 3000 N/m, $k_3$ = 1000 N/m

The system is assumed to be:

• Undamped

• Linear

• Free vibration (no external force)

## Methodology

The natural frequencies are determined using the following computational approach:

Define mass and stiffness values

Construct mass matrix [M]

Construct stiffness matrix [K]

Solve generalized eigenvalue problem using MATLAB

Compute $\omega = \sqrt{\lambda}$

Convert rad/s to Hz

Plot natural frequency graph

Plot vibration time response

## MATLAB Implementation

The MATLAB script performs:

• Definition of system parameters

• Formation of mass and stiffness matrices

• Eigenvalue solution using eig(K,M)

• Sorting natural frequencies

• Displaying numerical results

• Plotting:

Natural frequency vs Mode number

Time response vibration curve

The built-in eigenvalue solver is used for accurate computation.

→ Results

The computational analysis yields:

Natural Frequencies:

• $\omega_1 \approx 21.73$ rad/s

• $\omega_2 \approx 45.98$ rad/s

In Hz:

• $f_1 \approx 3.46$ Hz

• $f_2 \approx 7.32$ Hz

## Graphical Output

The program generates:

• Natural Frequency vs Mode Number graph

• Smooth sinusoidal vibration response curve

• Mode shape representation

These graphs help visualize dynamic characteristics of the system.

## APPLICATION:-

Natural frequency estimation is essential in:

• Vehicle suspension design

• Machine foundation analysis

• Rotor critical speed determination

• Bridge vibration analysis

• Aerospace structural design

• Earthquake-resistant structures

Avoiding resonance ensures safety, reliability, and structural stability.

This project highlights the importance of matrix methods and computational tools in modern mechanical engineering vibration analysis.

## How to Run the Project

- Open MATLAB
- Copy the provided .m file
- Run the script

Observe: • Natural frequencies in Command Window

• Frequency vs Mode graph

• Time response vibration curve

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Computational Vibration Analysis Project

## Project Summary

This project successfully demonstrates how natural frequencies of a spring–mass system can be determined using matrix eigenvalue methods in MATLAB.

It reinforces the importance of vibration analysis, computational modeling, and eigenvalue techniques in modern engineering system design.

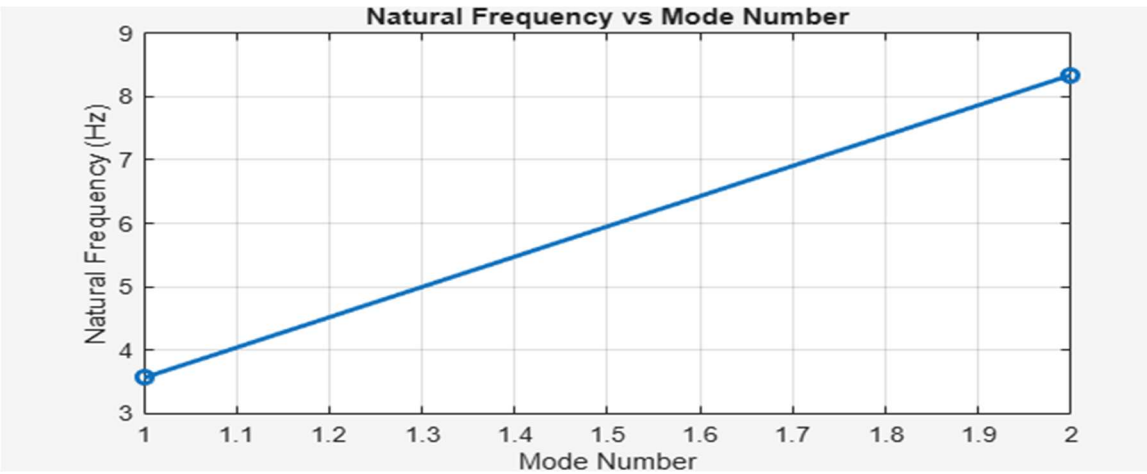
## MATLAB CODE:-

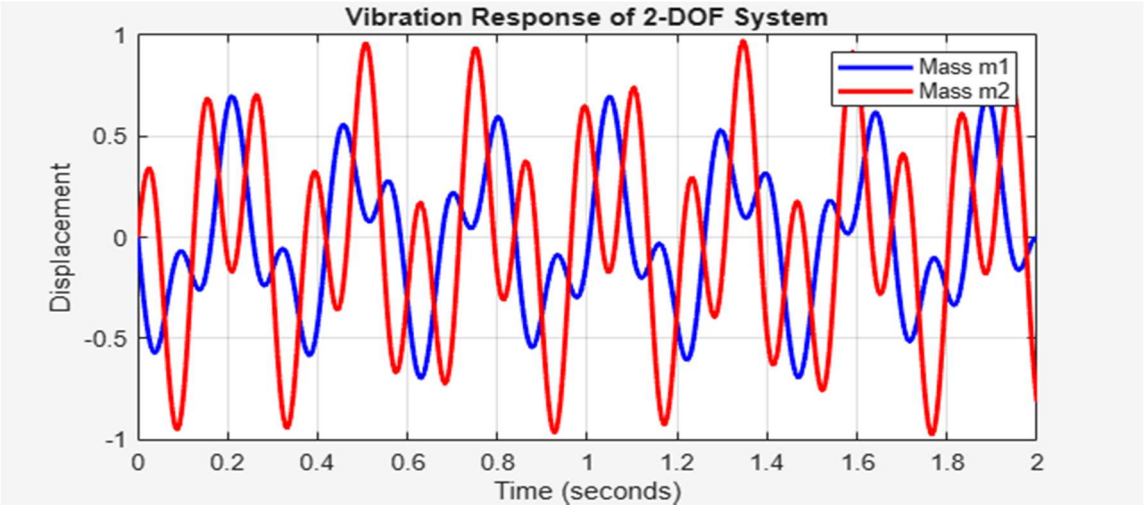
```
clc;
clear;
close all;
% Given Example Values

m1 = 4;      % kg
m2 = 2;      % kg

k1 = 2000;   % N/m
k2 = 3000;   % N/m
k3 = 1000;   % N/m
% Mass Matrix [M]
M = [m1 0;
     0  m2];
% Stiffness Matrix [K]

K = [k1+k2  -k2;
     -k2    k2+k3];
% Solve Eigenvalue Problem

[V,D] = eig(K,M);

lambda = diag(D);          % Eigenvalues
omega = sqrt(lambda);      % Natural frequency (rad/s)
f = omega/(2*pi);          % Natural frequency (Hz)

% Sort values (Mode 1 -> Mode 2)
[omega, idx] = sort(omega);
f = f(idx);
V = V(:,idx);

% Display Results

disp('================================');
disp('Natural Frequencies (rad/s):');
disp(omega);

disp('Natural Frequencies (Hz):');
disp(f);
disp('================================');

% Expected approx:
% omega1 ≈ 21.73 rad/s
% omega2 ≈ 45.98 rad/s
```

```
% f1 ≈ 3.46 Hz
% f2 ≈ 7.32 Hz
% GRAPH 1: Natural Frequency vs Mode Number
figure;
plot(1:2, f, 'o-', 'LineWidth', 2);
grid on;
xlabel('Mode Number');
ylabel('Natural Frequency (Hz)');
title('Natural Frequency vs Mode Number');

% GRAPH 2: Smooth Vibration Curve (Time Response)
% x(t) = Phi*sin(wt)
t = 0:0.001:2;   % Time (seconds)
x1 = V(1,1)*sin(omega(1)*t) + V(1,2)*sin(omega(2)*t);
x2 = V(2,1)*sin(omega(1)*t) + V(2,2)*sin(omega(2)*t);
figure;
plot(t, x1, 'b', 'LineWidth', 2);
hold on;
plot(t, x2, 'r', 'LineWidth', 2);
grid on;

xlabel('Time (seconds)');
ylabel('Displacement');
title('Vibration Response of 2-DOF System');
legend('Mass m1','Mass m2');
```

## Result :-

```
Natural Frequencies (rad/s):
   22.3607
   52.4404

Natural Frequencies (Hz):
    3.5588
    8.3462
```




Vibration Response of 2-DOF System
Mass m1
Mass m2
Displacement
Time (seconds)
1
0.5
0
-0.5
-1
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2